King Saud University
Muzahmyiah branch
College of Applied computer science

Subject: ***Network Simulation***

ID course: ***COS 372***

Term: 2026 FALL

project title: VLAN and InterVLAN

project no: 2

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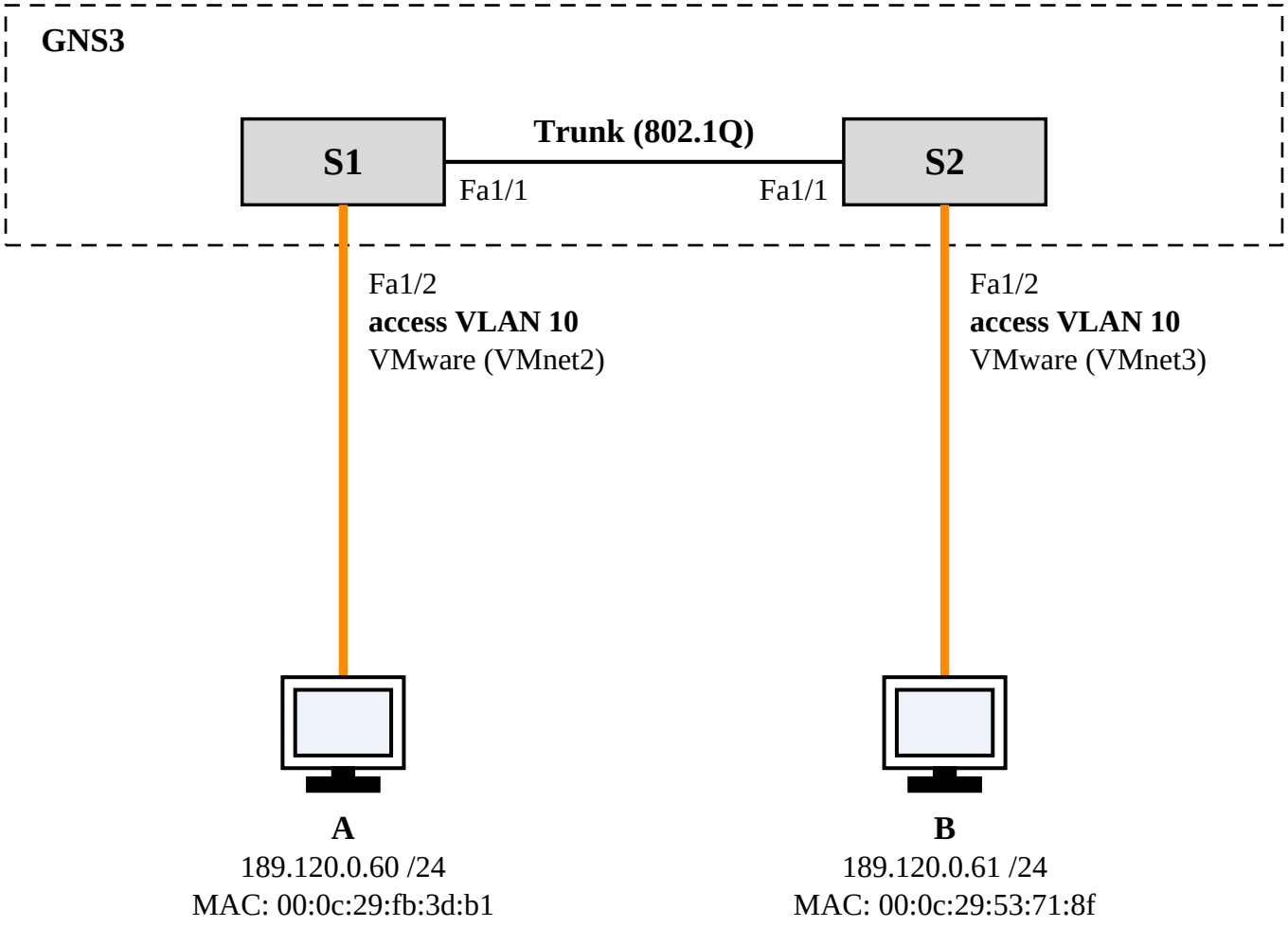
VLAN & InterVLAN Routing

VLAN

Ping from A to B, both in VLAN 10 (no router)

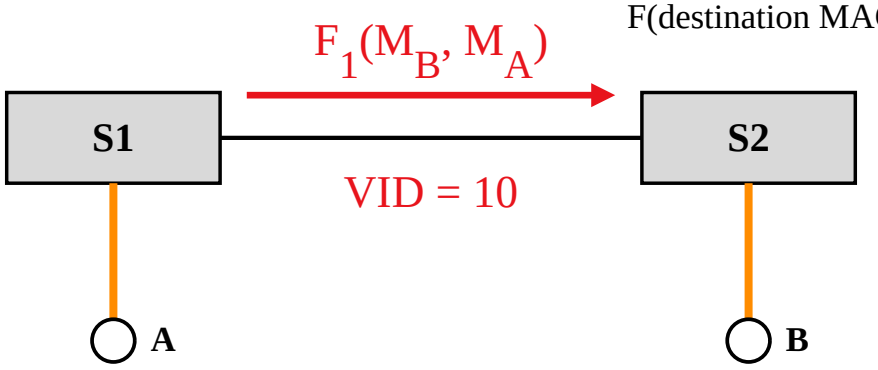
TOPOLOGY



ECHO REQUEST

A → B

F(destination MAC, source MAC)



$F_1(M_B, M_A)$ VID = 10 Echo Request from A to B (78 bytes):

```
0000   00 0c 29 53 71 8f 00 0c   29 fb 3d b1 81 00 00 0a
0010   08 00 45 00 00 3c 03 1f   00 00 80 01 bc 38 bd 78
0020   00 3c bd 78 00 3d 08 00   4c e0 00 01 00 7b 61 62
0030   63 64 65 66 67 68 69 6a   6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63   64 65 66 67 68 69
```

00	0C	29	53	71	8F	00	0C	29	FB	3D	B1	81	00	00	0A
08	00	45	00	00	3C	03	1F	00	00	80	01	BC	38	BD	78
00	3C	BD	78	00	3D	08	00	4C	E0	00	01	00	7B	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From A to B (VLAN 10) →

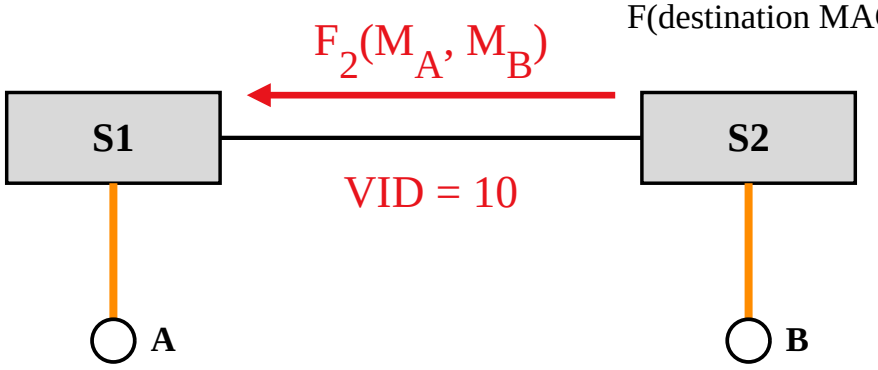
DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
00 0C 29 53 71 8F	00 0C 29 FB 3D B1	81 00 00 0A	08 00	TTL = 128	08	
6 bytes (B)	6 bytes (A)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0000 1010
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 10*

ECHO REPLY

B ← A

F(destination MAC, source MAC)



$F_2(M_A, M_B)$ VID = 10 Echo Reply from B to A (78 bytes):

```
0000   00 0c 29 fb 3d b1 00 0c   29 53 71 8f 81 00 00 0a
0010   08 00 45 00 00 3c 03 f5   00 00 80 01 bb 62 bd 78
0020   00 3d bd 78 00 3c 00 00   54 e0 00 01 00 7b 61 62
0030   63 64 65 66 67 68 69 6a   6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63   64 65 66 67 68 69
```

00	0C	29	FB	3D	B1	00	0C	29	53	71	8F	81	00	00	0A
08	00	45	00	00	3C	03	F5	00	00	80	01	BB	62	BD	78
00	3D	BD	78	00	3C	00	00	54	E0	00	01	00	7B	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From B to A (VLAN 10)

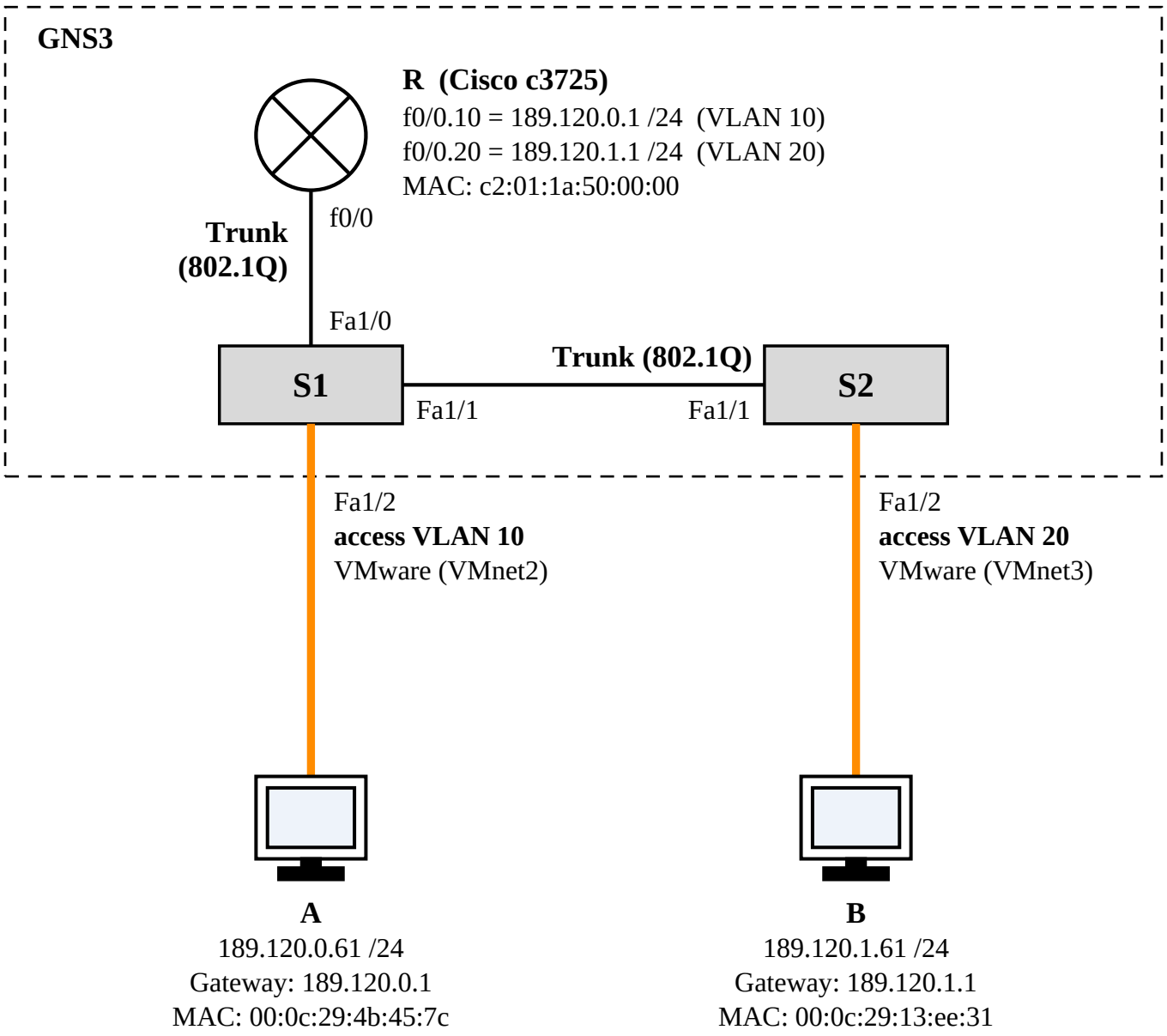
DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
00 0C 29 FB 3D B1	00 0C 29 53 71 8F	81 00 00 0A	08 00	TTL = 128	00	
6 bytes (A)	6 bytes (B)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0000 1010
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 10*

InterVLAN

Router on a stick: ping from A (VLAN 10) to B (VLAN 20) through the router R

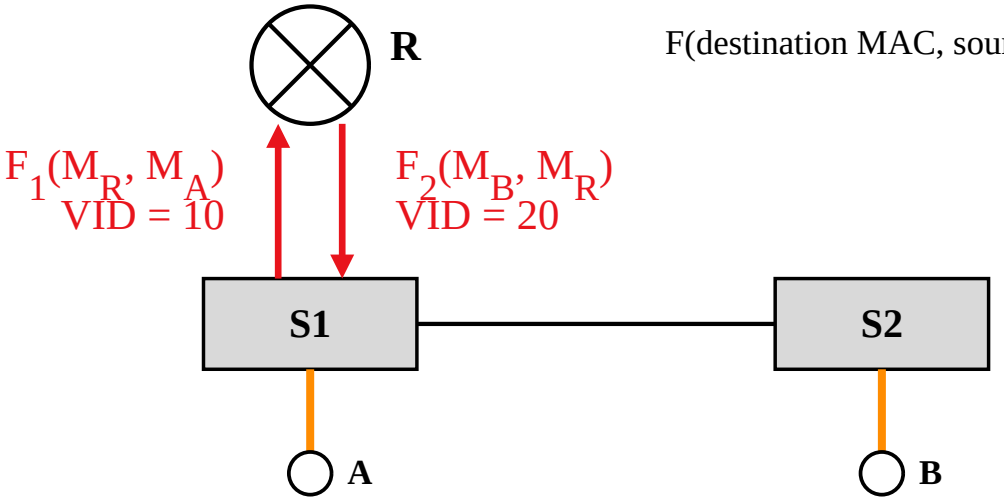
TOPOLOGY



ECHO REQUEST

A → R → B

F(destination MAC, source MAC)



$F_1(M_R, M_A)$ VID = 10 Echo Request from A to R (78 bytes):

```
0000   c2 01 1a 50 00 00 00 0c  29 4b 45 7c 81 00 00 0a
0010   08 00 45 00 00 3c 00 9a  00 00 80 01 bd bc bd 78
0020   00 3d bd 78 01 3d 08 00  4d 56 00 01 00 05 61 62
0030   63 64 65 66 67 68 69 6a  6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63  64 65 66 67 68 69
```

C2	01	1A	50	00	00	00	0C	29	4B	45	7C	81	00	00	0A
08	00	45	00	00	3C	00	9A	00	00	80	01	BD	BC	BD	78
00	3D	BD	78	01	3D	08	00	4D	56	00	01	00	05	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From A to R (VLAN 10) →

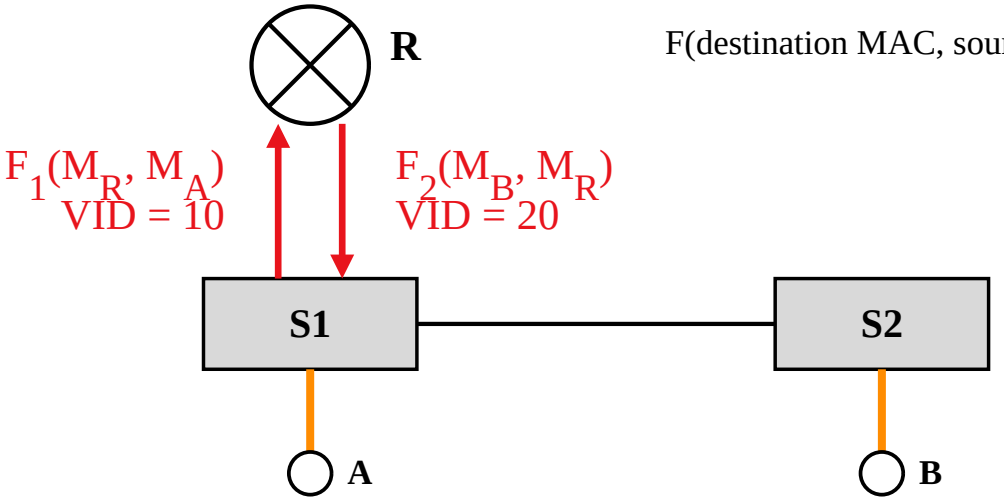
DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
C2 01 1A 50 00 00	00 0C 29 4B 45 7C	81 00 00 0A	08 00	TTL = 128	08	
6 bytes (R)	6 bytes (A)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0000 1010
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 10*

ECHO REQUEST

A → R → B

F(destination MAC, source MAC)



$F_2(M_B, M_R)$ VID = 20 Echo Request from R to B (78 bytes):

```
0000   00 0c 29 13 ee 31 c2 01  1a 50 00 00 81 00 00 14
0010   08 00 45 00 00 3c 00 9a  00 00 7f 01 be bc bd 78
0020   00 3d bd 78 01 3d 08 00  4d 56 00 01 00 05 61 62
0030   63 64 65 66 67 68 69 6a  6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63  64 65 66 67 68 69
```

00	0C	29	13	EE	31	C2	01	1A	50	00	00	81	00	00	14
08	00	45	00	00	3C	00	9A	00	00	7F	01	BE	BC	BD	78
00	3D	BD	78	01	3D	08	00	4D	56	00	01	00	05	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From R to B (VLAN 20) →

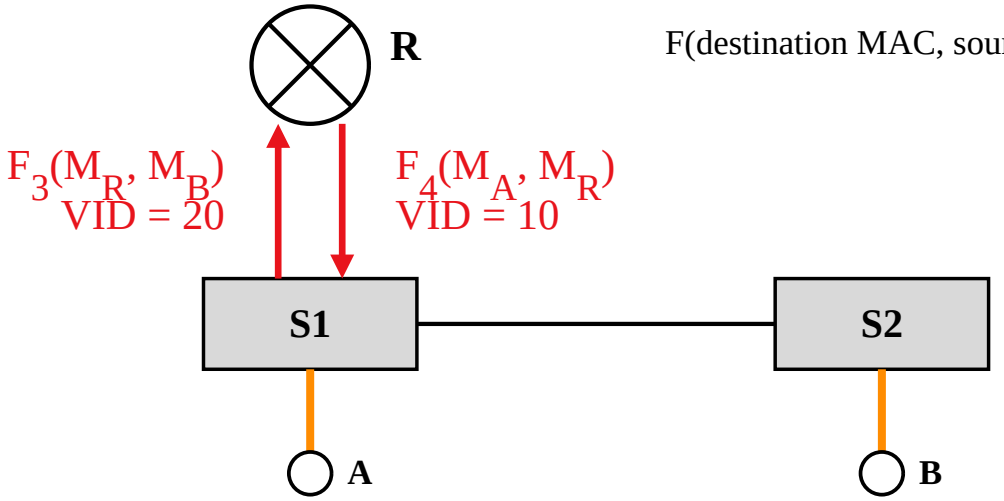
DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
00 0C 29 13 EE 31	C2 01 1A 50 00 00	81 00 00 14	08 00	TTL = 127	08	
6 bytes (B)	6 bytes (R)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0001 0100
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 20*

ECHO REPLY

B → R → A

F(destination MAC, source MAC)



$F_3(M_R, M_B)$ VID = 20 Echo Reply from B to R (78 bytes):

```
0000   c2 01 1a 50 00 00 00 0c  29 13 ee 31 81 00 00 14
0010   08 00 45 00 00 3c 00 56  00 00 80 01 be 00 bd 78
0020   01 3d bd 78 00 3d 00 00  55 56 00 01 00 05 61 62
0030   63 64 65 66 67 68 69 6a  6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63  64 65 66 67 68 69
```

C2	01	1A	50	00	00	00	0C	29	13	EE	31	81	00	00	14
08	00	45	00	00	3C	00	56	00	00	80	01	BE	00	BD	78
01	3D	BD	78	00	3D	00	00	55	56	00	01	00	05	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From B to R (VLAN 20)

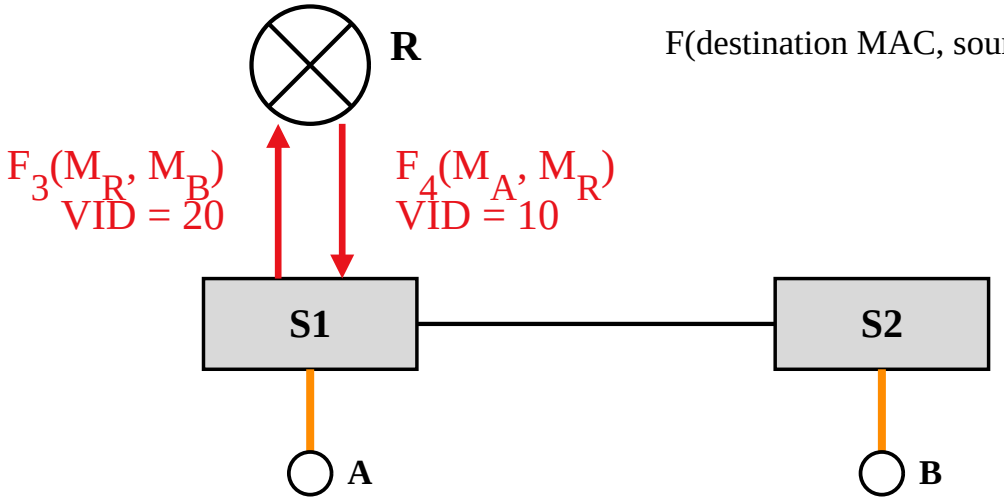
DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
C2 01 1A 50 00 00	00 0C 29 13 EE 31	81 00 00 14	08 00	TTL = 128	00	
6 bytes (R)	6 bytes (B)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0001 0100
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 20*

ECHO REPLY

B → R → A

F(destination MAC, source MAC)



$F_4(M_A, M_R)$ VID = 10 Echo Reply from R to A (78 bytes):

```
0000   00 0c 29 4b 45 7c c2 01   1a 50 00 00 81 00 00 0a
0010   08 00 45 00 00 3c 00 56   00 00 7f 01 bf 00 bd 78
0020   01 3d bd 78 00 3d 00 00   55 56 00 01 00 05 61 62
0030   63 64 65 66 67 68 69 6a   6b 6c 6d 6e 6f 70 71 72
0040   73 74 75 76 77 61 62 63   64 65 66 67 68 69
```

00	0C	29	4B	45	7C	C2	01	1A	50	00	00	81	00	00	0A
08	00	45	00	00	3C	00	56	00	00	7F	01	BF	00	BD	78
01	3D	BD	78	00	3D	00	00	55	56	00	01	00	05	61	62
63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72
73	74	75	76	77	61	62	63	64	65	66	67	68	69		

From R to A (VLAN 10)

DEST MAC	SOURCE MAC	802.1Q TAG	TYPE	IP HEADER	ICMP	CRC
00 0C 29 4B 45 7C	C2 01 1A 50 00 00	81 00 00 0A	08 00	TTL = 127	00	
6 bytes (A)	6 bytes (R)	4 bytes	2 bytes	20 bytes	40 bytes	4 bytes

TPID (16 bits)	PCP (3 bits)	CFI (1 bit)	VID (12 bits)
81 00	000	0	0000 0000 1010
= 0x8100 (802.1Q)	*priority = 0*	*= 0*	*VLAN ID = 10*